BIOCHEMISTRY

1.nucleosome:

a) contain negatively charged protein
b) Is coiling of each single strand of DNA and histone
c) Is association of each DNA and arginine rich protein
d) Is combination of DNA and chromatin

2. **A palm of full-term baby showed single transverse palmer creases the chromosomal abnormality most likely to be?**
   a) 45 xy
   b) 47 xy +18
   c) xy +21
   d) 69 xxy

3. **Lactic acidosis caused by deficiency of?**
   a) thymine
   b) B1
   c) folic acid
   d) Biotin

4. **Which clinical laboratory observation below is suggestive of Hartnup disease (neutral amino acid transport deficiency)?**
   a) Burnt-sugar smell in urine
   b) High plasma phenylalanine levels
   c) Elevated plasma tyrosine and methionine levels
   d) High indole derivatives

5. **Beriberi?**
   a) not common in alcoholics
   b) the deficiency of riboflavin
   c) affect the nerves and the heart
   d) is the combination of diarrhea, dementia and Dermatitis

6. **Lipoprotein lipase:**
   a) Hydrolyses cholesteryl ester
   b) Is present in the cytosol of adipocytes

c) Is active during starvation
d) Hydrolyses TAG in chylomicrons or VLDL

7. Pyruvate carboxylase?

a) converts acetyl COA to pyruvate
b) require carnitine
c) converts pyruvate to oxaloacetate
d) is activated by high fructose 1,6 biphosphate

8. Synthesis of glucose from pyruvate by gluconeogenesis in the liver:

a) Occurs exclusively in the liver
b) Is inhibited during prolonged fasting
c) Requires participation of Biotin
d) Requires lactate as intermediate
e) Occurs exclusively in the mitochondria

9. Both glucogenic and ketogenic amino acids except:

a) phenylalanine
b) tyrosine
c) valine
d) threonine

10. in DNA Watson and crick model:

a) thymine base pairs with adenine
b) thymine base pairs with cytosine
c) DNA is a single strand
d) The purine content is not equal to the pyrimidine content
e) The nitrogenous bases from the backbone of the molecule

11. the bonds that stabilize the structure of DNA are:

a) hydrogen bonds and peptide ponds
b) glycosidic bonds and thioester bonds
c) phosphodiester bonds and hydrogen bonds
d) disulphide bonds and hydrogen bonds
e) Van dar Waals forces

12. why glutamate is important in transfer of ketoacid in trans deamination reaction;

a) alpha ketoglutarate is abundant in body

b) oxidative deamination in the liver

13. Cholesterol?

a) Normal cell membrane constituent
b) Precursor of sphingolipids
c) Main source of energy in the body
d) Produced by HMG lyase

14. synthetic nucleotide analogs:

a) Can inhibit DNA synthesis
b) Are always modified purine nucleotides
c) Lack phosphate group
d) Allow replication to proceed at faster rate

15. All of the following are consumed in urea cycle except:

a) aspartate
b) Co2
c) Ornithine
d) NH3

16. Which of the following statement about galactosemia is correct?

a) Treatment involves restriction of glucose in the diet
b) A deficiency of glucokinase is the major cause
c) Is treated by elimination of galactose from the diet
d) Can be treated by restricting sucrose in the diet
e) is inherited as autosomal dominant character

17. Arachidonic acid:

a) Is branched fatty acid
b) Contains 3 double bond
c) Is a precursor for eicosanoids
d) Contain 5 double bonds
e) Is a20-carbon saturated fatty acid

18. the following are purine nucleotides:

a) CTP– GDP
b) NAD– TDP
c) GMP – ADP
d) UDP– GTP

19. During prolonged fasting?

a) B-Oxidation of fatty acid is initiated
b) glycogenesis is very active
c) blood glucose level is maximum
d) ketogenesis is maximum

20. All the following use ketone bodies oxidation as significant source of energy except:

a) liver
b) kidney
c) RBC
d) heart

21. Which of the following amino acids is glucogenic but not Ketogenic;

a) Tyrosine
b) Tryptophan
c) Valine
d) Phenylalanine
e) Isoleucine

22. Which phase of meiosis we found DNA coupled and cross over:

a) Metaphase1
b) Metaphase 2
c) Anaphase
d) Prophase1
e) Telophase

23. In experiment a poison was injected and caused disruption to rough endoplasmic Reticulum. Which of the following will be affected:

a) Protein synthesis
b) DNA replication
c) Transcription
d) Transformation
e) Protein storage

24. Which of the following is true about DNA ligase:

a) it links DNA fragments
b) it acts in leading strands
c) act to recoil strands

d) act only in 5 to 3 direction
e) make template strand

25. Mutations which occur in body Cells which do not go on to form Gametes can be classified as:

a) auxotrophic mutations
b) somatic mutations
c) morphological mutations
d) oncogenes
e) temperature sensitive mutations

26. Polyploidy refers to:

a) extra copies of a gene adjacent to each other on a chromosome
b) an individual with complete extra sets of chromosomes
c) a chromosome which has replicated but not divided
d) multiple ribosomes present on a single mRNA
e) an inversion which does not include the centromere

27. A gene showing codominance.

a) Has both alleles independently Expressed in the heterozygote
b) has one allele dominant to the other
c) has alleles tightly linked on the same chromosome
d) has alleles expressed at the same time in development
e) has alleles that are recessive to each other

28. X-chromosome inactivation

a) normally takes place in males but not females
b) is the cause of the Y chromosome being genetically inactive
c) takes place in humans so that the same X chromosome is inactive in all of the cells of a female
d) occurs in fruit flies but not in mammals
e) results in genetically turning off one of the two X chromosomes in female mammals

29. The polymerase chain reaction or PCR is a technique that

a) was used to demonstrate DNA as the genetic material
b) is used to determine the content of Minerals in a soil sample
c) uses short DNA primers and Thermostable DNA Polymerase to replicate Specific DNA sequences in Vitro.

d) measures the ribosome transfer rate During translation Detects the level of polymerases Involved in replication

30. antimycin inhibit electron transport chain which of the following not affected.

a) Proximal convoluted tubule
b) RBC
c) Nerve cell
d) Cardiac cell

31. Replication of DNA

a) takes place in a “conservative” manner
b) takes place in a “dispersive” manner
c) takes place in a “semiconservative” manner
d) usually involves one origin of replication per chromosome in eukaryotes
e) takes place only in the 3’ to 5’ direction

32. Proteins are soluble in

a) anhydrous acetone.
b) aqueous alcohol
c) anhydrous alcohol
d) benzene

33. The hydrogen bond between peptide linkages of a protein molecules is interfered by

a) guanidine
b) uric acid
c) oxalic acid
d) salicylic acid

35. A lipoprotein inversely related to the incidence of coronary atherosclerosis is

a) VLDL
b) LDL
c) IDL
d) HDL

34. The amino acid with a nonpolar side chain is

a) Serine
b) Valine
c) Asparagine

d) Threonine

35. In case of repeated contraction, muscle cells regenerate NAD By converting pyruvate to

a) Glycogen
b) Lactate
c) Starch
d) Cellulose

36. what is the total gain of energy during anaerobic respiration

a) Two ATP
b) one ATP
c) Four ATP
d) Three ATP

37. Conversion of alanine to carbohydrate is termed

a) Glycogenesis
b) Gluconeogenesis
c) Glycogenolysis
d) Photosynthesis

38. The following is an enzyme required for glycolysis

a) Pyruvate kinase
b) Pyruvate carboxylase
c) Glucose -6-phosphatase
d) Glycerol kinase

39. A ketogenic amino acid is

a) B. valine
b) Cysteine
c) Leucine
d) Threonine

40. glycogen is converted to glucose-1-phosphate by

a) UDPG transferase
b) Branching enzyme
c) Phosphorylase
d) Phosphatase

41. Sucrose is composed of

a) Glucose

b) Glucose and Glycogen
c) Two molecules of glucose
d) Glycogen and fructose

42. The intermediate in hexose monophosphate shunt is

a) D-Ribulose
b) D-Arabinose
c) D-Xylose
d) D-lyxose

43. Which of the following is a mitochondrial enzyme that mediates all steps in the conversion of cholesterol to pregnenolone and rate limiting step for the entire steroid synthetic pathway

a) P450P10
b) P450scc
c) p450mwc
d) p450ACC

44. Two conditions in which gluconeogenesis is increased are

a) Diabetes mellitus and atherosclerosis
b) Fed condition and thyrotoxicosis
c) Diabetes mellitus and starvation
d) alcohol intake and cigarette smoking

45. Pyruvate dehydrogenase complex and ketoglutarate dehydrogenase complex require the following for their oxidative decarboxylation:

a) COASH and Lipoic acid
b) NAD+ and FAD
c) COASH and TPP
d) COASH, TPP, NAD, FAD, Lipoate.

46. High energy molecule needed to convert oxaloacetate to phosphoenoyl pyruvate

a) GTP
b) ITP
c) ITP and GTP
d) None of the above

47. Monosaccharides are

a) glucose and fructose
b) aldose and hexose

c) maltose

48. **Ehlers –Danlos syndrome characterized by hypermobile joints and skin is due to**
    a) Abnormality in gene for pro collagen
    b) Deficiency of lysyl oxidase
    c) Deficiency of prolyl hydroxylase
    d) Deficiency of lysyl hydroxylase

49. **If you examine primordial germ cells in a prepuberal male under the microscope, in which stage of cell cycle will you find them?**
    a) Prophase
    b) Anaphase
    c) Telophase
    d) Interphase

50. **Which of the following is a true statement in an otherwise normal gestation?**
    a) in the presence of SRY, a female phenotype will develop
    b) in the absence of SRY, a female phenotype will develop
    c) SRY is located on the X chromosome
    d) Only one X chromosome is necessary for normal ovarian development.

51. **20 years old girl attends the gynecology clinic with her mother, she presents with primary amenorrhea, on examination she is tall with BMI 19Kg/m2 Blind end vagina. there is no axillary or pubic hair. What is the most likely diagnosis?**
    a) Complete androgen insensitivity syndrome
    b) Klinefelter syndrome
    c) Mullerian agenesis
    d) Turner syndrome

52. **M.T. is a 27-year-old white woman who has been attempting pregnancy for 15 months. she is monitoring her ovulation with LH testing. her test today indicated that she will ovulate tomorrow. in what phase will the ovulated germ cell be arrested?**
    a) Prophase of first mitotic division
    b) Anaphase of first meiotic division
    c) Metaphase of second meiotic division
    d) Zygotic phase

53. **pregnant lady carrier for gene responsible for Duchenne muscular dystrophy her children will**
    a) 50% of sons and 50 % of daughters will be affected
    b) 50% of sons will be affected
    c) 100% of sons will be affected
    d) 100% of daughters will be affected.

54. A study involving analysis of risk for genetic defects in a family is:

a) Genetic Engineering
b) Genetics counselling
c) Genetics drift
d) Genetics equilibrium

55. In ELISA technique, the antibodies are labelled by:

a) Acridine orange
b) Alkaline phosphate
c) Neutral red
d) Bromothymol blue

56. Reverse transcriptase is an enzyme involved in the synthesis of:

a) DNA
b) Soluble RNA
c) m-RNA from DNA
d) Nucleotides

57. Phenotype is determined by:

a) sex chromosome (XX- XY) karyotyping
b) gonadal differentiation
c) steroid hormones

58. Poly A tail is frequently found in:

a) Histone in RNA
b) Bacterial RNA
c) Eukaryotic RNA
d) tRNA

59. The technique involved in comparing the DNA component of two sample is known as:

a) monoclonal Ab technique
b) genetic finger printing
c) recombinant DNA technology
d) PCR

60. If you want to produce different combination of traits:

a. gene recombination b.

61. Which of the following produces pyruvate?

a) Leucine
b) Isoleucine

c) Lysine
d) Alanine

62. The enzyme required for DNA from RNA template:

a) RNA polymerase
b) Reverse transcriptase
c) DNA polymerase
d) Terminal transferase

63. The enzyme responsible for the removal of supercoiling in replicating DNA ahead of the replication fork is:

a) Topoisomerase
b) Primase
c) DNA polymerase
d) Helicase

64. Which effects the increased levels of hydrogen ions in the inter-membrane space of the mitochondria?

a) Decreased levels of chemiosmosis
b) Increased levels of water in inter-membrane space
c) Increase ATP production
d) Decreased levels of oxidative phosphorylation

65. Which of the following enzyme is secreted by the pancreas?

a) Myoglobin
b) Cytochrome c
c) Ribonuclease
d) Lysozyme

66. Which compounds are responsible for the coordinated regulation of glucose and glycogen metabolism?

a) Fructose 2, 6-bisphosphate
b) Acetyl Co-A
c) NADH
d) NAD+

67. Which of the following is a measure of how responsive the enzyme is to changes in the concentration of a metabolite?

a) Metabolic control
b) Response coefficient
c) Flux control coefficient
d) Elasticity coefficient

68. Which of the following are exclusively glucogenic?

a) Lysine
b) Leucine
c) Threonine

d) Arginine

69. Which is the first step in the catabolism of most L-amino acids once they have reached the liver is promoted?

a) Amino transferases
b) Glutaminase
c) Glutamine synthase
d) Glutamate dehydrogenase

70. Which of the following operates at an important intersection of carbon and nitrogen metabolism?

a) Amino transferases
b) Glutaminase
c) Glutamine synthase
d) Glutamate dehydrogenase

71. Which of the following plays a substantial role in linking together sister chromatids immediately after replication?

a) Topoisomerases
b) Histones
c) Condensins
d) Cohesins

72. Which of the following is an example of RNA-dependent DNA polymerase?

a) RNA polymerase II
b) DNA ligase
c) Reverse transcriptase
d) RNA polymerase I

74. Which is the first amino group entering into urea cycle?

a) carbamoyl phosphate
b) Ornithine
c) Cituilline
d) Argininosuccinate

75. Family pedigree, skipped generation from male-to-male transmission, female is affected as equal as male, with heterozygous affected and homozygous is carrier?!

a) Autosomal dominant
b) Autosomal recessive
c) X-linked dominant
d) X- linked recessive

76. The primary structure of a protein

a) is determined by the sequence of amino acids

b) is a result of folding in the amino acid chain
c) is overseen by molecular chaperones such as heat shock proteins
d) has no effect on the functionality of the final molecule

77. guestion about aneuploidy:

78. In b12 deficiency which of the following found in urine:

a) Malate
b) Methylmalonate
c) Pyruvate
d) Lactate

79. The 4 rate limiting enzymes of gluconeogenesis are:

a) Glucokinase, Pyruvate carboxylase, phosphoenolpyruvate carboxykinase and glucose-6-phosphatase
b) Pyruvate carboxylase, phosphoenolpyruvate carboxykinase, fructose 1,6 bisphosphatase and glucose-6-phosphatase
c) Pyruvate kinase, pyruvate carboxylase, phosphoenolpyruvate carboxykinase and glucose-6-phosphatase
d) Phosphofructokinase, pyruvate carboxylase, phosphoenolpyruvate carboxykinase and fructose 1, 6 bisphosphatase

80. Ethanol decreases gluconeogenesis by:

a) Inhibiting glucose-6-phosphatase
b) Inhibiting PEP carboxykinase
c) Converting NAD+ into NADH and decreasing the availability of pyruvate
d) Converting NAD+ into NADH and decreasing the availability of lactate

81. Compared to the resting state, vigorously contracting muscle shows:

a) An increased conversion of pyruvate to lactate
b) Decreased oxidation of pyruvate of CO2 and water
c) A decreased NADH/NAD+ ratio
d) Decreased concentration of AM

82. Which one of the following would be expected in pyruvate kinase deficiency?

a) Increased levels of lactate in the R.B.C
b) Haemolytic anaemia
c) Decreased ratio of ADP to ATP in R.B.C
d) Increased phosphorylation of Glucose to Glucose-6-phosphate

83. Which one of the following statements concerning glucose metabolism is correct?

a) The conversion of Glucose to lactate occurs only in the R.B.C
b) Glucose enters most cells by a mechanism in which Na+ and glucose are co-transported
c) Pyruvate kinase catalyses an irreversible reaction
d) An elevated level of insulin leads to a decreased level of fructose 2, 6-bisphosphate in hepatocyte

84. Enzymes catalysing electron transport are present mainly in the:

a) Ribosomes
b) Endoplasmic reticulum
c) Lysosomes
d) Inner mitochondrial membrane

85. Mature erythrocytes do not contain:

a) Glycolytic enzymes
b) HMP shunt enzymes
c) Pyridine nucleotide
d) ATP

86. In mammalian cells rRNA is produced mainly in the:

a) Endoplasmic reticulum
b) Ribosome
c) Nucleolus
d) Nucleus

87. Inherited deficiency of β−β−β−glucosidase causes:

a) Tay-Sachs disease
b) Metachromatic leukodystrophy
c) Gaucher's disease

88. Which one of the following statements concerning glucose-6-phosphate dehydrogenase deficiency is correct?

a) Young RBCs, particularly reticulocytes, contain the highest enzyme activity cells show less enzyme activity
b) Glucose-6-P Dehydrogenase deficiency leads to dysfunction of many tissues
c) G-6-p Dehydrogenase deficiency is due to a single deletion of a large sequence of DNA in the G-6-PD gene
d) G-6-PD deficiency is precipitated by ingestion of drugs such as aspirin

89. Glycogen branching after how many glucose units?

a) 50
b) 10
c) 15
d) 20

90. Which of the following statements is correct about membrane cholesterol?

a) The hydroxyl group is located near the centre of the lipid layer
b) Most of the cholesterol is in the form of a cholesterol ester
c) The steroid nucleus form forms a rigid, planar structure
d) The hydrocarbon chain of cholesterol projects into the extracellular fluid

91. The enzymes which are commonly used in genetic engineering are:

a) Exonuclease and ligase
b) Restriction endonuclease and polymerase
c) Ligase and polymerase
d) Restriction endonuclease and ligase

92. The smallest unit of genetic material that can undergo mutation is called:

a) Gene
b) Cistron
c) Replicon
d) Muton

93. DNA replication:

a) M phase
b) S phase
c) G phase
d) G0 phase

﴿وَمَا تَوْفِيقِي إِلَّا بِاللَّهِ ۚ عَلَيْهِ تَوَكَّلْتُ وَإِلَيْهِ أُنِيبُ﴾